



Data mining final project

Customer Behavior & Music

Recommendation

Daniel Coblentz

Agenda

- Introduction
- Best Classification Model: Customer Purchase Prediction
- Best Clustering Algorithm: Music Recommendation
- Conclusion & Recommendations
- Q&A
- References

Classification Project Overview & Data Preprocessing:

(Project PT₁: Customer Behavior Classification (CBC Dataset))

- **Goal:**
Predict whether a customer will purchase *"The Art History of Florence"* (target variable: **Florence**).
- **Key Preprocessing Steps:**
 - **Feature Selection:** Dropped irrelevant columns *Seq#* and *ID#*.
 - **Missing Values:** No missing values were found.
 - **Outlier Handling:** Outliers were identified using IQR and Z-score methods but retained as they represent real customer behavior. Logistic Regression and Random Forest models are robust to moderate outliers aswell.
 - **Data Splitting:** Dataset split into 60% training, 20% validation, and 20% testing sets to ensure robust model evaluation.
 - **Feature Scaling:** Numeric features were standardized using StandardScaler to prevent features with larger magnitudes from dominating model training.
 - **Class Imbalance:** The target variable *Florence* was highly imbalanced (91.55% non-purchasers vs. 8.45% purchasers). This was addressed using `class_weight='balanced'` in the Logistic Regression model.

Best Classification Model — Logistic Regression

Model Selection Rationale:

- Logistic Regression demonstrated the best ROC AUC Score among all evaluated models.
- It provided a good balance between identifying the minority class and overall discriminative ability.

Model Configuration:

- **Algorithm:** Logistic Regression
- **Parameters:** penalty='l2', C=1.0, solver='lbfgs', max_iter=5000
- **Class Imbalance Handling:** class_weight='balanced' was applied to account for the imbalanced target classes.
- **Performance Comparison (AUC):**
 - Logistic Regression (Baseline): **0.6384**
 - Logistic Regression + SMOTE: **0.6361**
 - Random Forest (Baseline): **0.6224**
 - Random Forest + SMOTE: **0.5893**

Classification Model Results & Evaluation(Baseline)

	precision	recall	f1-score	support
0	0.94	0.66	0.78	729
1	0.14	0.56	0.22	71
accuracy			0.65	800
macro avg	0.54	0.61	0.50	800
weighted avg	0.87	0.65	0.73	800

Classification Model Results: Logistic Regression (Baseline)

Key Metrics (Test Set):

- **Accuracy:** 0.6538
- **ROC AUC Score:** 0.6384

Classification Model Results & Evaluation(SMOTE)

	precision	recall	f1-score	support
0	0.94	0.66	0.78	729
1	0.13	0.54	0.21	71
accuracy			0.65	800
macro avg	0.54	0.60	0.50	800
weighted avg	0.87	0.65	0.73	800

Classification Model Results: Logistic Regression (Baseline)

Key Metrics (Test Set):

- **Accuracy:** 0.6525
- **ROC AUC Score:** 0.6361

Baseline vs. SMOTE-upsampled Logistic Regression

The baseline Logistic Regression slightly outperformed the SMOTE-upsampled version.

- **AUC:** Baseline (0.6384) vs. SMOTE (0.6361)
- **Accuracy:** Baseline (0.6538) vs. SMOTE (0.6525)

Clustering Project Overview & Data Preprocessing

(Project PT2: Spotify Music Recommendation System)

Cluster music tracks based on their audio features to build a content-based recommendation system.

Key Preprocessing Steps:

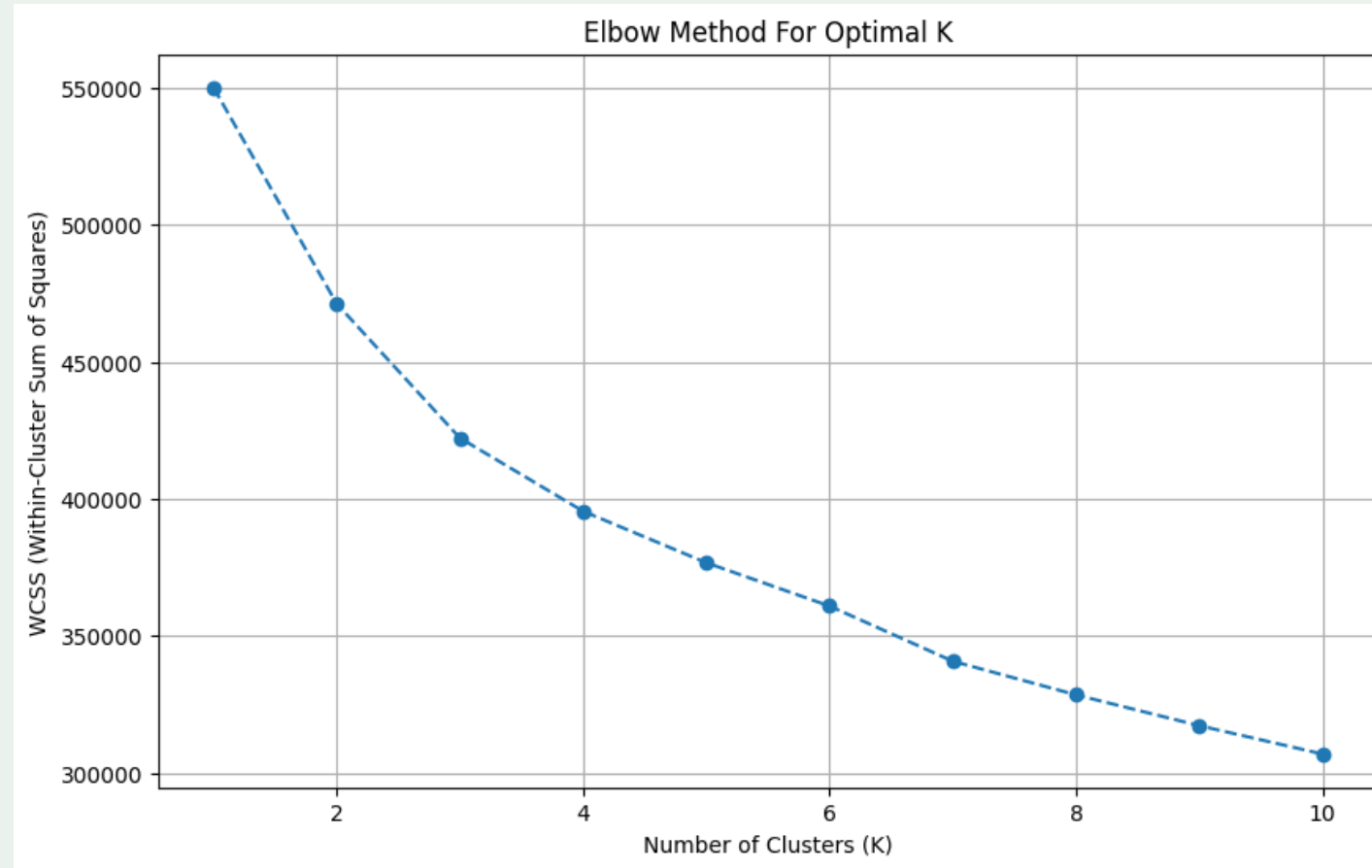
- **Feature Selection:** Dropped non-numeric and irrelevant identifier columns such as type, id, uri, track_href, analysis_url, song_name, title, and Unnamed: 0.
- **Genre Handling:** The genre column was stored separately for potential interpretation but removed from the feature set used for clustering.
- **Missing Values:** No missing values were found in the selected numeric features.
- **Feature Scaling:** All numeric audio features were standardized using StandardScaler to ensure equal influence during distance calculations.

Best Clustering Algorithm — K-Means with Optimal K

Algorithms Evaluated:

- K-Means Clustering
- Agglomerative Hierarchical Clustering
- Elbow method for finding optimal K value

Both clustering methods got the same K value -->



Clustering Algorithm Results & Interpretation

Algo) Kmeans clustering results + PCA plot

K-Means Silhouette Score: 0.1487

K-Means Davies-Bouldin Index: 2.1681

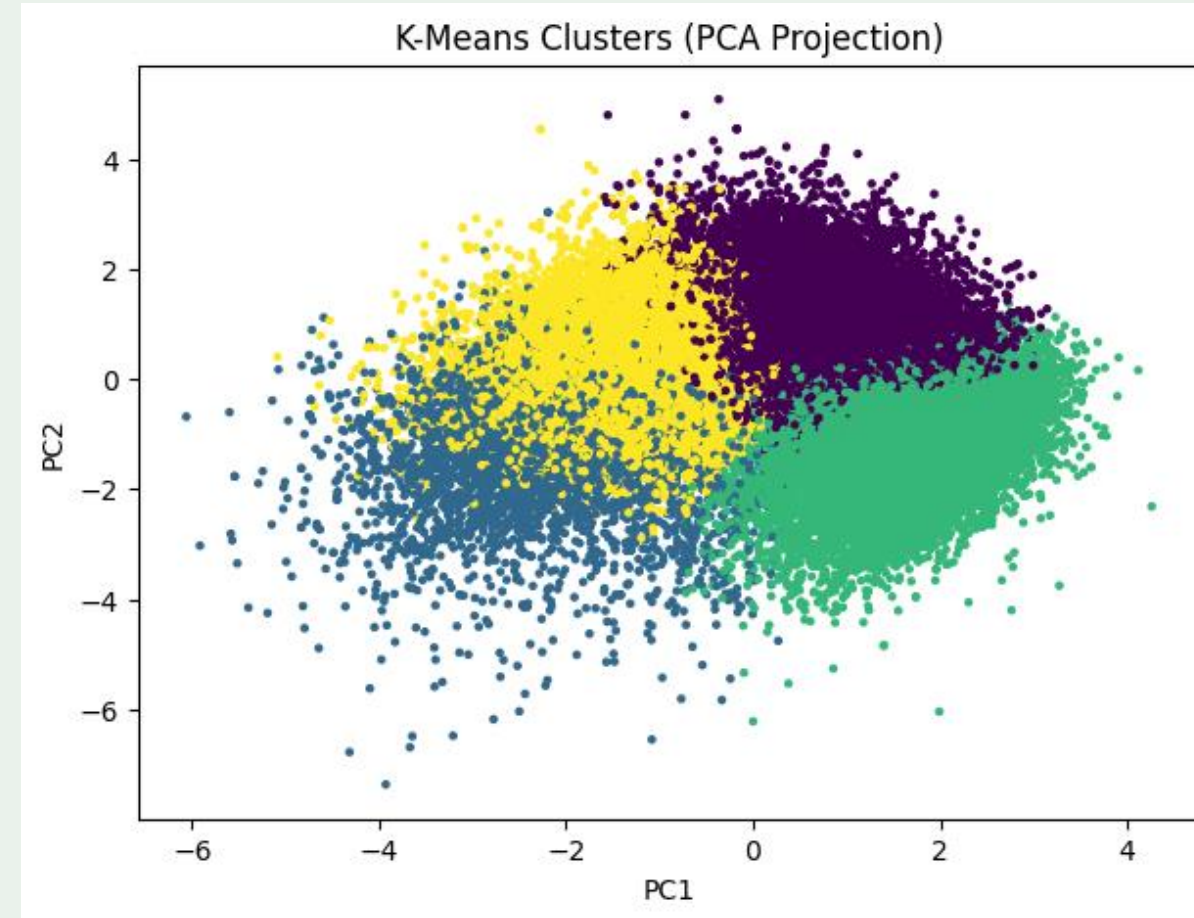
Cluster table:

C0 (Purple) --> High danceability, Moderate energy, Moderate speechiness,

C1 (Green) --> high energy, High liveness, Longer duration

C2 (Yellow) --> high instrumentalness, low speechiness & acousticness

C3 (Blue) --> experimental styles, Progressive rock, jazz fusion



Summary of results

Aspect	Agglomerative (k=4)	K-means (k=4)
Silhouette Score	0.1227	0.1487 (higher, better separation)
Cluster Count	4 broad clusters	4 detailed clusters
Interpretability	Simple: Vocal/Danceable vs Instrumental/Intense	More nuanced: live, ambient, complex rhythms
Visualization (PCA)	More overlapping clusters	Clearer, more compact separation
Best For	Foundational recommendation system	Niche, fine-grained playlist systems

Conclusion & Recommendations

Project PT1: Customer Behavior Classification

- **Best Model:** Logistic Regression with `class_weight='balanced'`.
- **Key Outcome:** Achieved an AUC of 0.6384, providing the best discriminative power for predicting Florence purchases.
- **Recommendation:** Implement this model to identify potential high-value customers for targeted marketing campaigns. Further analysis of false positives/negatives can refine targeting.

Project PT2: Spotify Music Recommendation System

- **Best Model:** K-Means Clustering with `k=4`.
- **Key Outcome:** Achieved a Silhouette Score of 0.1487, indicating two reasonably well-separated clusters based on acoustic features.
- **Recommendation:** Use these clusters as the foundation of a music recommendation system. Further exploration can support playlist generation and preference modeling.

References

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- [3] MacQueen, J. (1967). Some methods for classification and analysis of multivariate observations. *Proceedings of the fifth Berkeley symposium on mathematical statistics and probability*, 1(14), 281-297.
- [4] Rousseeuw, P. J. (1987). Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20, 53-65.
- [5] Davies, D. L., & Bouldin, D. W. (1979). A cluster separation measure. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, PAMI-1(2), 224-227.